

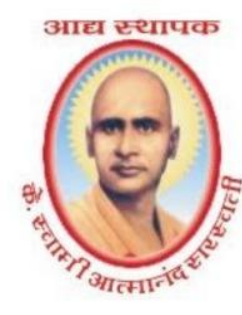


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TAPI DIPLOMA ENGINEERING COLLEGE

Formerly : Shree Tapi Brahmcharyashram Sabha College of Diploma Engineering

Approved by AICTE New Delhi & Affiliated to GTU Ahmedabad

Accredited by N.B.A. (Mechanical & Computer)
(National Board of Accreditation, New Delhi.)



ONLINE VOTING & ELECTION MANAGEMENT SYSTEM

Minor Project – DI05000341 | Seminar - 1

College Detail

Tapi Diploma Engineering College
Computer Department

Group Detail

Prepared By : All Group Members

Subject Detail

Minor Project
Seminar Presentation - 1

Vision: "To mould technocrat in the field of computer engineering with innovation skills, moral values and societal concerns."

Presentation Index

As per Minor Project Seminar-1 instruction



- 1 Introduction
- 2 Literature Survey
- 3 Problem Identification
- 4 Proposed Solution & Work Plan
- 5 Tools and Technologies
- 6 Expected Outcomes
- 7 References

Introduction

Project Title and background of the domain



Project Title

Online Voting and Election Management System
A digital platform for voter registration, authentication, secure voting, and result management.

Domain Background

Election management involves voter verification, ballot preparation, vote casting, counting, and result publishing.
A web-based system can reduce manual work and improve transparency.

Core Idea

Provide a secure and easy online process where authorized voters can vote once and administrators can manage elections efficiently.

Need of the Project

Problems in existing systems and benefits of the proposed system



Problems in Current Manual System

Long queues and time-consuming voter verification process.

Manual counting may create delay and human error.

Difficult to manage records and election reports quickly.

Voters may face location and accessibility barriers.

Benefits of Proposed System

Faster election process with digital voter authentication.

One voter, one vote control using secure login and OTP.

Automatic counting and quick result generation.

Better record management and admin monitoring dashboard.

Objectives

Main goals of the system



Secure Access

Authenticate voters using login credentials and OTP verification.

Accurate Voting

Allow each verified voter to vote only once.

Admin Control

Manage voters, candidates, elections, and results from a dashboard.

Fast Results

Generate automatic counting reports and result summaries.

Data Protection

Store vote data securely and maintain privacy of voter choice.

User Friendly

Provide simple screens for voters and election officers.

Scope and Applications

Where the system can be used



College Elections

Student council, class representative, club elections.

Society Elections

Housing society committee or association voting.

Organization Polls

Internal employee voting and policy decisions.

Local Community Voting

Community-level decision making and surveys.

Scope includes voter registration, candidate profile, election scheduling, vote casting, result dashboard, and report generation.

Literature Survey

Review of existing systems, tools and key findings



Existing System / Product	Tools / Technology Used	Key Finding
Traditional paper ballot	Manual voter list and counting	Simple but slow and difficult to audit quickly.
Electronic voting machines	Hardware-based voting device	Fast counting but needs physical polling booth setup.
Online survey tools	Web forms and databases	Easy to use but not built for strong election security.
Research e-voting systems	Encryption, authentication, audit trail	Security and privacy are the major focus areas.

Problem Identification

Limitations, research gap and problem statement



Limitations of Existing System

Physical presence is required at a fixed location.

Counting and record preparation may be delayed.

Risk of duplicate or invalid voting if verification is weak.

Research Gap

Need for a simple online system for small-scale elections.

Need to balance easy user experience with secure authentication.

Need for quick reports and transparent admin control.

Problem Statement

To develop a secure, user-friendly online voting and election management system that reduces manual work, prevents duplicate voting, and generates reliable election results.

Proposed Solution

Concept of the proposed system



The system will provide two main modules: Voter Module and Admin Module.

Voter Module

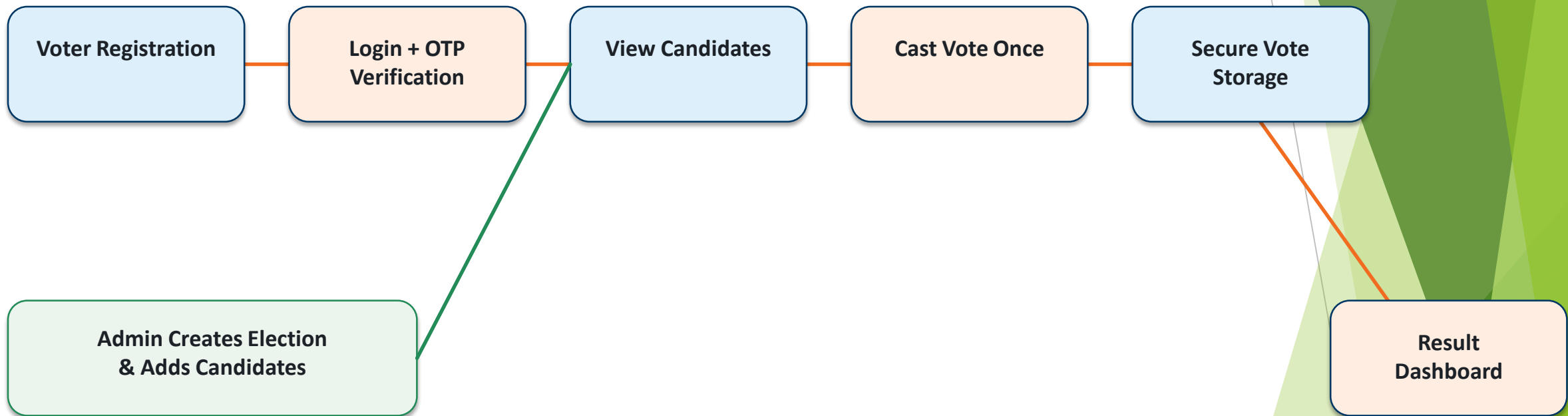
- Register / Login
- OTP-based verification
- View candidate list
- Cast vote once
- View election status

Admin Module

- Add and manage elections
- Add voters and candidates
- Monitor voting status
- Generate result dashboard
- Export reports

Methodology / Flow Diagram

Process flow of the online voting system



Vote is accepted only after authentication and duplicate-vote checking.

Tools and Technologies

Frontend and backend plan



Frontend

HTML5
CSS3 / Bootstrap
JavaScript

Backend

PHP / Node.js
REST API
Server-side validation

Database

MySQL / MongoDB
Voter data
Election results

Security

OTP verification
Password hashing
Role-based access

Tools

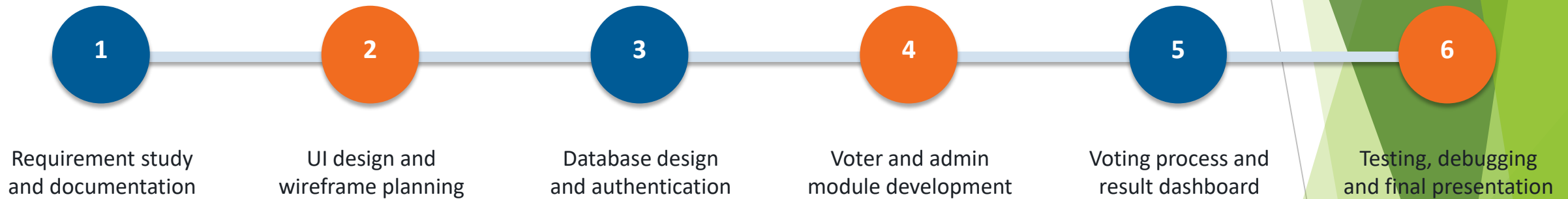
VS Code
XAMPP / Node environment
Git & GitHub

Testing

Functional testing
Security testing
User acceptance testing

Work Plan and Timeline

Expected timeline for project completion



Timeline can be adjusted according to college schedule and project review feedback.

Expected Outcomes

Deliverables and expected project result



Working Prototype

A functional online voting web application.

Secure Authentication

Voter login, OTP verification, and duplicate voting prevention.

Admin Dashboard

Election, voter, candidate, and result management.

Result Reports

Automatic vote counting and summary generation.

Documentation

Project report, flow diagram, testing and references.

Future Scope

Blockchain-based audit trail and biometric verification can be added later.

References

Authentic websites, research sources and standards



References Used for Study

Election Commission of India - official information on elections and voting process.

NIST Digital Identity Guidelines - authentication and identity proofing concepts.

OWASP Web Security Testing Guide - web application security testing practices.

IEEE Xplore / Google Scholar - research papers on electronic voting and secure e-voting systems.

MDN Web Docs - HTML, CSS and JavaScript documentation.

MySQL / MongoDB Documentation - database design and query reference.

THANK YOU